

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO)

Q.1 Solve Any Two of the following.

- A) Define with example with suitable examples 6
i) Subsets ii) Power set iii) NULL Set iv) Universal set.
- B) Draw the Venn diagrams for each of these combinations 6
of the sets A, B, and C.
a) $A \cap (B - C)$ b) $(A \cap B) \cup (A \cap C)$
c) $(A \cap \overline{B}) \cup (A \cap \overline{C})$
- C) 6
In a class of 55 students, the number of students studying different subjects are 23 in Mathematics and 24 in Physics, 19 in Chemistry, 12 in Mathematics and Physics, 9 in Mathematics and Chemistry, 7 in Physics and Chemistry and 4 in all the three subjects, Find the number of students who have taken exactly one subject?

Q.2 Solve Any Two of the following.

- A) Let g be the function from the set {a, b, c} to itself such that $g(a) = b$, $g(b) = c$, and $g(c) = a$. Let f be the function from the set {a, b, c} to the set {1, 2, 3} such that $f(a) = 3$, $f(b) = 2$, and $f(c) = 1$. What is the composition of f and g, and what is the composition of g and f? 6
- B) Let $X = \{x, y, z, k\}$ and $Y = \{1, 2, 3, 4\}$. Determine which of the following are functions. Give reasons if it is not. Find range if it is a function. 6
1. $f = \{(x, 1), (y, 2), (z, 3), (k, 4)\}$
2. $g = \{(x, 1), (y, 1), (k, 4)\}$
3. $h = \{(x, 1), (x, 2), (x, 3), (x, 4)\}$
4. $d = \{(x, 1), (y, 2), (y, 3), (z, 4), (z, 4)\}$
- C) Two mathematics papers & 5 other papers are to be arranged at an examination 6

Find the total no of ways if Math papers are consecutive.

Q. 3 Solve Any Two of the following.

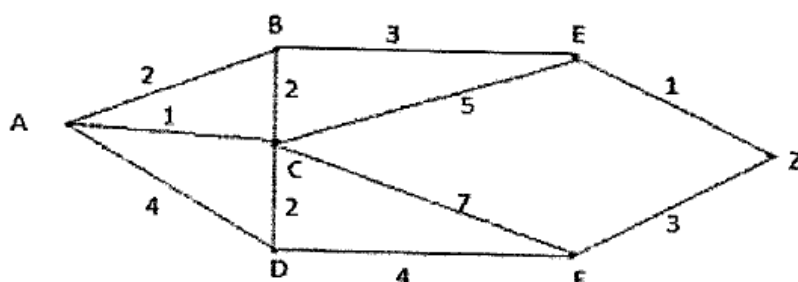
- A) Define with example: 1.Injective(one-to-one) function 2.Surjective(Onto) function 3.Bijective(one-to-one onto) function 6
- B) Find the number of distinct permutations that can be formed from all the letters of each word. I) RADAR II) UNUSUAL III) COMBINATION IV) BANANA 6
- C) Prove using mathematical induction that for all $n \geq 1$, 6
 $1 + 4 + 7 + \dots + (3n - 2) = n(3n - 1) / 2$

Q.4 Solve Any Two of the following.

- A) Define i) Integral Domain ii) Commutative Ring with unity iii) Field 6
- B) Consider these relations on the set of integers: 6
 $R1 = \{ (a,b) \mid a \leq b \}$
 $R2 = \{ (a,b) \mid a > b \}$
 $R3 = \{ (a,b) \mid a = b \text{ or } a = -b \}$
 $R4 = \{ (a,b) \mid a = b \}$
 $R5 = \{ (a,b) \mid a = b + 1 \}$
 $R6 = \{ (a,b) \mid a + b \leq 3 \}$
Which of these relations are transitive?
- C) Explain applications of database and relations? 6

Q. 5 Solve Any two of the following.

- A) Define with example: 1.Null Graph 2.Undirected Graph 3.Multigraph 6
- B) Draw Binary Search Tree for input data 3, 1,4,6,9,2,5,7 6
List out a root node, leaf nodes and interior nodes.
- C) Apply Dijkstra's shortest path algorithm to find shortest path between vertices a and z in following graph. 6



*** End ***